

Markscheme

Mathematics: analysis and approaches

Practice question bank

150. (a)

Consider $z = -4 + 4i$.

Write z in the form $re^{i\theta}$, giving r exactly and θ (in radians) correct to 3 significant figures.

$$r = \sqrt{16 + 16} = 4\sqrt{2} \quad (M1)$$

$$\approx 5.66 \quad \mathbf{A1}$$

$$\theta = \pi - \arctan(1) \quad (M1)$$

$$\approx 2.36 \quad \mathbf{A1}$$

[4 marks]